

Finite-Blocklength Latency Minimization for Uplink and Downlink Communications: Per-User Scheduling Results

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Abstract

This document focuses only on per-user scheduling behavior. Throughout, the *Training Only Method* is used as the benchmark for the *Training + Testing Method*. The clearest scheduling gap appears in the downlink. In payload completion, the benchmark finishes with block counts (4, 3, 3), whereas the *Training + Testing Method* expands that to (9, 3, 7). In the uplink, the scheduling structures are already close, and the main visible difference is limited to a single extra payload block for user 1 under the *Training + Testing Method*.

1 Scope and Naming

This document is intentionally schedule-only. The terminology is aligned with the companion theory report:

- **Training Only Method:** direct online optimization during scheduling,
- **Training + Testing Method:** a separate training stage followed by testing-stage inference,
- **Payload completion:** the scenario where each user continues until its remaining payload is exhausted.

The *Training Only Method* is treated as the benchmark, and the *Training + Testing Method* is evaluated against it.

2 Downlink

The downlink schedule results separate the two methods most clearly. The main difference is not only how long the schedule is, but how efficiently service is distributed across users once multiuser coupling becomes difficult.

2.1 Payload Completion Results and Comparison

Downlink payload completion shows the strongest benchmark advantage. The *Training Only Method* finishes with user block counts (4, 3, 3), while the *Training + Testing Method* expands the same task to (9, 3, 7). That means the main schedule tail is carried by users 0 and 2.

Training Only Method

- **User 0:** 4 blocks, total $n = 198$, served 1000/1000, with 3 skipped blocks.
- **User 1:** 3 blocks, total $n = 140$, served 2000/2000, with no skipped blocks.
- **User 2:** 3 blocks, total $n = 199$, served 3000/3000, with no skipped blocks.

Training + Testing Method

- **User 0:** 9 blocks, total $n = 427$, served 1000/1000, with 7 skipped blocks.
- **User 1:** 3 blocks, total $n = 164$, served 2000/2000, with no skipped blocks.
- **User 2:** 7 blocks, total $n = 512$, served 3000/3000, with 3 skipped blocks.

Figure 1 makes the same point visually. The benchmark remains compact, with average chosen sub-blocklengths approximately (49.5, 46.7, 66.3). The *Training + Testing Method* pushes users 1 and 2 to heavier average sub-blocklengths, approximately (47.4, 54.7, 73.1), while also fragmenting users 0 and 2 across more blocks. So the remaining gap is not just a small timing shift; it is a materially longer and less synchronized schedule.

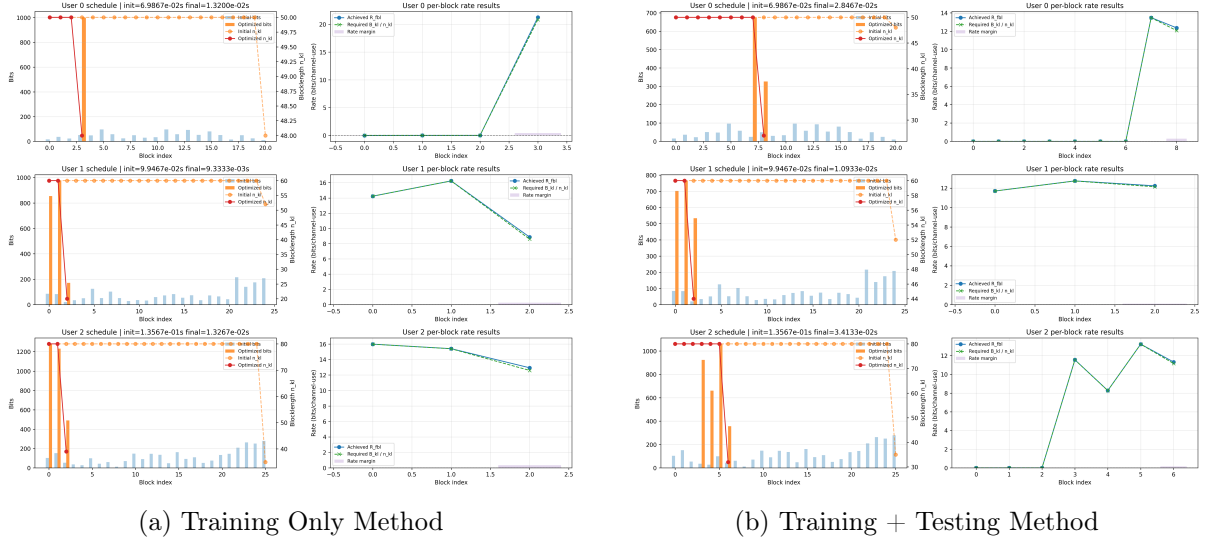


Figure 1: Downlink payload-completion schedule comparison. The benchmark keeps the schedule compact, while the *Training + Testing Method* stretches users 0 and 2 over a much longer block tail.

3 Uplink

The uplink results are much closer to the benchmark. The *Training + Testing Method* largely preserves the same per-user schedule shape and differs mainly through a small blocklength redistribution.

3.1 Payload Completion Results and Comparison

Uplink payload completion keeps the same overall front-loaded structure in both methods: most service is delivered early, and the last block is used to clear a small remainder. The benchmark finishes with block counts (2, 2, 3), while the *Training + Testing Method* gives (2, 3, 3). The only structural difference is the extra block for user 1.

Training Only Method

- **User 0:** 2 blocks, total $n = 68$, served 1000/1000, and cleared in 2 blocks.
- **User 1:** 2 blocks, total $n = 119$, served 2000/2000, and cleared in 2 blocks.
- **User 2:** 3 blocks, total $n = 183$, served 3000/3000, with a small final clean-up block.

Training + Testing Method

- **User 0:** 2 blocks, total $n = 67$, served 1000/1000, and cleared in 2 blocks.
- **User 1:** 3 blocks, total $n = 129$, served 2000/2000, with one extra block versus the benchmark.
- **User 2:** 3 blocks, total $n = 182$, served 3000/3000, with a small final clean-up block.

So the benchmark edge is very localized. Users 0 and 2 are essentially matched, while user 1 is the only case where the *Training + Testing Method* extends the schedule. Figure 2 shows that both methods still keep the same basic policy of a large early allocation followed by a sharply reduced final block.

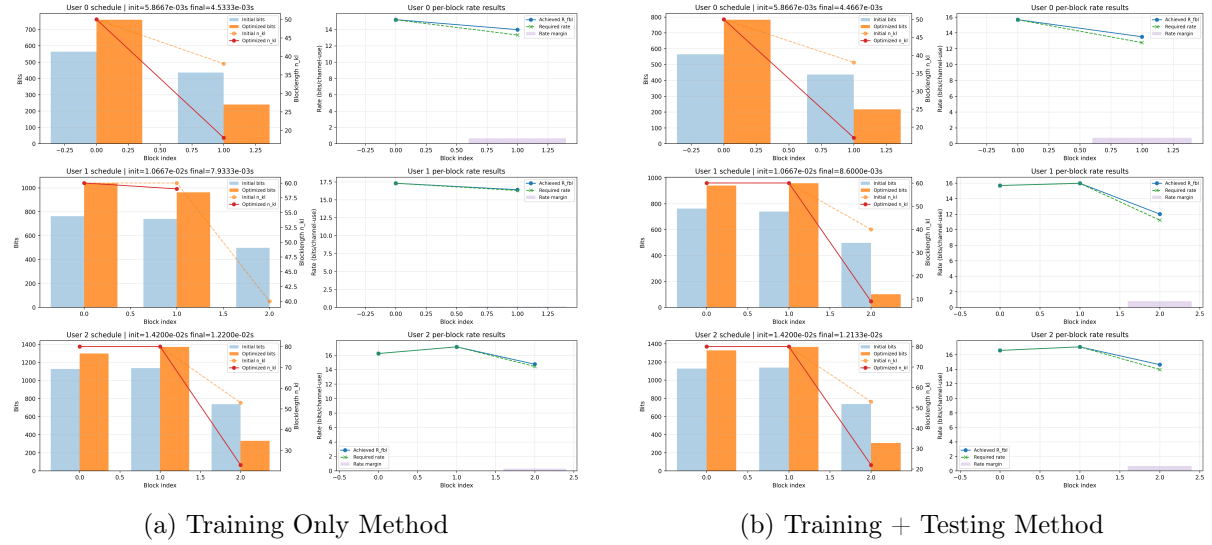


Figure 2: Uplink payload-completion schedule comparison. The two methods follow almost the same structure, with the only material change being one extra block for user 1 under the *Training + Testing Method*.

4 Schedule Takeaways

The scheduling results are most naturally read benchmark-first. In downlink payload completion, the main shortfall of the *Training + Testing Method* is schedule fragmentation, especially on users 0 and 2.

In the uplink, the schedule gap is already small. Payload completion differs mainly through one extra user-1 block. So the present scheduling gap is mostly a downlink issue, while the uplink is already close to the benchmark at the per-user schedule level.